α-controlled initial accessibility under unfrozen dynamics: circuits track $R(t)$, and global decay drags both below the boundary

converged-$d2$ pretrain, $V_1 \to \alpha V_1$ at $d4$ onset, everything trainable — 8 α × 8 λ × 3 seeds

Design

Same 4-bit, full-cube, two-block ReLU setup (N=1024, init var $1/N$, exact Walsh). Phase 1 pretrains the whole network on the parents to machine precision ($\lambda = 0$, 10^4 steps). At $d4$ onset the block-1 weights are rescaled $V_1 \to \alpha V_1$, $\alpha \in \{0.1, \ldots, 4\}$, setting the *measured* initial accessibility $R_0 = Q_2/Q_1 \in [0.04, 0.58]$ on both sides of $R_c = 1/\sqrt{8} \approx 0.354$ — the unfrozen analogue of the frozen α-sweep. Phase 2 trains *everything* ($d2$ losses kept, fresh optimizer) under a global weight decay $\lambda \in \{0, 10^{-4}, \ldots, 10^{-1}\}$ applied to all four matrices, identical initial state across λ within (seed, α). $R(t)$ and the pre-activation–based circuit fractions are recorded every 500 steps. Every cell solves $d4$ ($|\hat{f}_4| \geq 0.85$; no failures).

Findings

1. No compositional phase forms — initial accessibility biases, but does not select. Even from $R_0 = 0.58$, well above R_c, the final parent share only reaches $f_{2+2} = 0.10$ ($\lambda = 0$) or 0.15 ($\lambda = 10^{-4}$), and the $f_{2+2}(R_0)$ curves (Fig. 2, left) rise *smoothly* through R_c with no transition — in stark contrast to the frozen case, where the same R values give a near-step to $f_{2+2} \approx 0.9$. For $\lambda \geq 3{\cdot}10^{-3}$, $f_{2+2} = 0$ everywhere (Fig. 1).

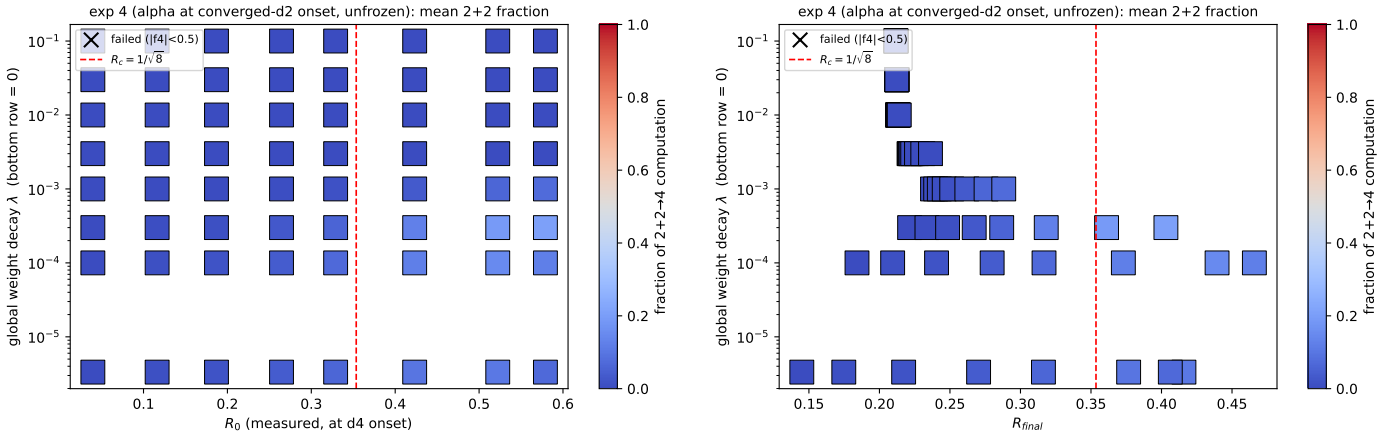


Figure 1: Phase diagrams vs. R_0 (left) and R_{final} (right), same color scale as the frozen-case diagram: uniformly in the direct (1^4) phase; the mild light-blue gradient at $\lambda \leq 10^{-4}$ is the residual imprint of R_0.

2. The drift direction depends on the initial value — until decay takes over. The drift map (Fig. 2) shows R_{final} vs. R_0: at $\lambda = 0$ the curve crosses the diagonal near $R_0 \approx 0.28$ — representations drift *up* from below and *down* from above, toward a broad band — and supercritical starts *stay* supercritical (R_{final} up to 0.47) while still building a mostly-direct circuit. For $\lambda \geq 10^{-3}$ the curves flatten: the decay fixed point $R^* \approx 0.21$–0.29 swallows every initial condition, erasing α entirely.

3. The circuit tracks $R(t)$, with no hysteresis. The phase plane (Fig. 3) shows why the final states are direct: at moderate decay ($\lambda = 3{\cdot}10^{-3}$), supercritical starts first *build* a compositional component — f_{2+2} transiently peaks at ≈ 0.23 while $R(t)$ slides left — and the moment $R(t)$ crosses R_c the 2+2 circuit is dismantled and rebuilt as 1^4, ending at $f_{2+2} = 0$, $R \approx 0.23$. At $\lambda = 0$ trajectories barely move: circuits form within the first few hundred steps near R_0 and freeze there. Selection is governed by the *instantaneous* accessibility during formation, not by R_0 or R_{final} alone.

4. Where the static picture survives. At $\lambda \leq 10^{-4}$ the unfrozen points land near the frozen curve when compared at R_{final} (e.g. unfrozen $f_{2+2} = 0.10$ at $R_{\text{final}} = 0.41$ vs. frozen 0.125 at $R = 0.38$) — an approximate collapse. At larger λ the comparison becomes ill-posed: R moves substantially *during* circuit formation, and only the trajectory explains the outcome. The frozen experiments' sharp R_c transition is thus a property of *pinned* representations; unfrozen dynamics soften it from above (drift) and below (early circuit lock-in).

Artifacts: `gdrive:parity/analysis_output/accessibility_alpha_unfrozen/` (`results_merged.csv`, per-slice results/trajectories, figures, this report); code: `analysis/accessibility/run_alpha_unfrozen_experiment.py`, `merge_alpha_unfrozen.py`. Companions: `accessibility_report.pdf`, `accessibility_wd_report.pdf`, `accessibility_unfrozen_report.pdf`. This experiment doubles as the t_{pre}=10^4 column of the running t_{pre} sweep (exp. 5, norway/texas).

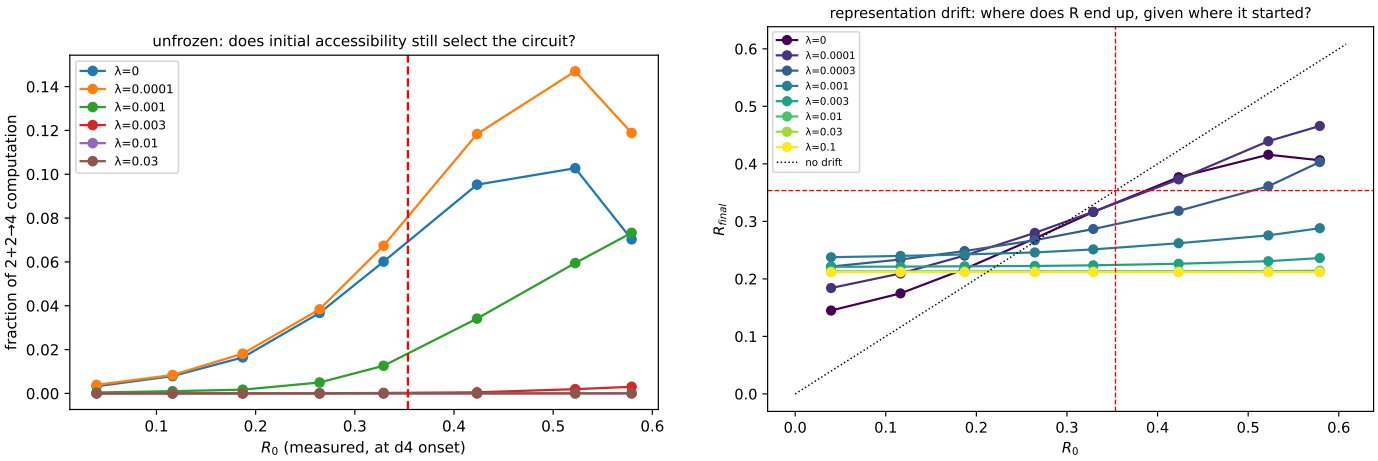


Figure 2: Left: f_{2+2} vs. R_0 at fixed λ — smooth growth through R_c, no transition, maximum 0.15. Right: representation drift $R_{\text{final}}(R_0)$ per λ (dotted: no drift; red: R_c). At $\lambda = 0$ the curve crosses the diagonal near 0.28 (bidirectional drift; supercritical starts stay supercritical); for $\lambda \geq 10^{-3}$ a λ-set fixed point $R^* \approx 0.21$–0.29 erases the initial condition.

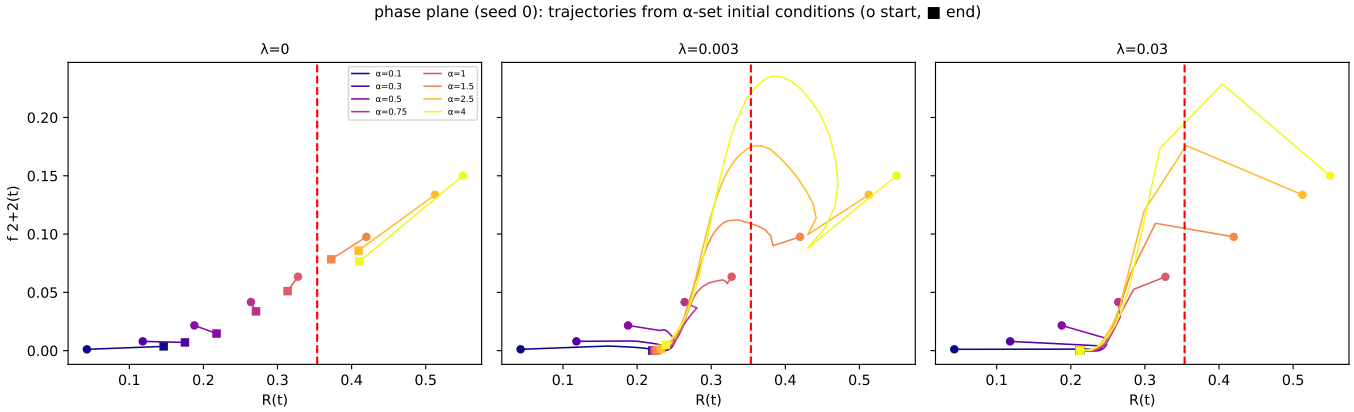


Figure 3: Phase-plane trajectories $R(t)$ vs. $f_{2+2}(t)$ (seed 0; ∘ start, ■ end; color = α). Left, $\lambda = 0$: near-static — circuits lock in close to R_0. Middle/right, $\lambda = 3{\cdot}10^{-3}/3{\cdot}10^{-2}$: supercritical starts grow a transient 2+2 fraction, then lose it as decay pulls $R(t)$ across R_c; all end direct at R^*.